

Solutions to Exercise sheet 1

Exercise 1.1 Sum of Cubes (1 point).

Prove by mathematical induction that for every positive integer n ,

$$1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}.$$

Solution

Base case: $n=1$

$$1^3 = 1 = \frac{1^2(1+1)^2}{4} = \frac{1 \cdot 4}{4}$$

$\Rightarrow$ The statement holds $n=1$

Induction Hypothesis: We assume the statement holds for $n=k$, $k \in \mathbb{N}$

$$\Rightarrow 1^3 + 2^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4}$$

Induction Step: We will prove the statement holds for $n=k+1$

We want to show that: $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \frac{(k+1)^2((k+1)+1)^2}{4}$

$$\begin{aligned} & \underbrace{1^3 + 2^3 + \dots + k^3}_{(*)} + (k+1)^3 = \\ & \frac{k^2(k+1)^2}{4} + (k+1)^3 = \end{aligned}$$

(From I.H. we know $1^3 + 2^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4}$)

$$\frac{k^2(k+1)^2}{4} + \frac{4(k+1)^3}{4} =$$

$$\frac{k^2(k+1)^2 + 4(k+1)(k+1)^2}{4} =$$

$$\frac{(k+1)^2(k^2 + 4(k+1))}{4} =$$

$$\frac{(k+1)^2(k^2 + 4k + 4)}{4} =$$

$$\frac{(k+1)^2(k+2)^2}{4} =$$

$$\frac{(k+1)^2((k+1)+1)^2}{4}$$

$$\Rightarrow \text{We showed } 1^3 + 2^3 + \dots + k^3 + (k+1)^3 = \frac{(k+1)^2((k+1)+1)^2}{4}$$

$\Rightarrow$ we have shown that the statement holds for $n=k+1$

$\Rightarrow$ By the principle of the mathematical induction $1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$ holds for any positive integer n .

! Don't forget the final conclusion

Comment: Master solution does the same calculations but instead of doing (*) in the beginning takes the assumption from the hypothesis and adds $(k+1)^3$ to both sides.

This solution demonstrates an appropriate level of detail and formalism for the exam.

Exercise 1.2 Sum of reciprocals of roots (1 point).

Consider the following claim:

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \leq \sqrt{n}.$$

A student provides the following induction proof. Is it correct? If not, explain where the mistake is.

Base case: $n = 1$,

$$\frac{1}{\sqrt{1}} \leq 1, \text{ which is true.}$$

Induction hypothesis: Assume the claim holds for $n = k$, i.e.

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k}.$$

Induction step: Then, starting from the claim we need to prove for $n = k + 1$ and using logical equivalences:

$$\begin{aligned} \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1} &\iff \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k+1} - \frac{1}{\sqrt{k+1}} \\ &\iff \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k+1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+1}} \\ &\iff \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \leq \frac{k}{\sqrt{k}} \leq \sqrt{k}, \end{aligned}$$

which is true, therefore the claim holds by the principle of mathematical induction.

Solution

We notice that the claim does not hold for $n = 2$

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} > \sqrt{2}.$$

However, the proof by induction shows that the claim holds for every $n \in \mathbb{N} \Rightarrow$ the proof is incorrect.

The Base case and the Induction hypothesis are both correct $\Rightarrow$ the error must be in the Induction step.

Let us consider the logical equivalence

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1} \Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k+1} - \frac{1}{\sqrt{k+1}} \quad (\text{correct})$$

$$\Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k+1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+1}} \quad (\text{correct})$$

$$\Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \quad (\text{until here everything is correct})$$

From here the solution of the student continues by saying

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \leq \frac{k}{\sqrt{k}} \leq \sqrt{k} \text{ as part of the equivalence}$$

from where he uses the fact that we have assumed

$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k}$ in the induction hypothesis with the goal to show that his claim ($\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1}$) is equivalent to something that we know is true.

$\Rightarrow$ The correctness of the student's solution gets reduced to the correctness of the equivalence

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1} \Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \leq \frac{k}{\sqrt{k}} \leq \sqrt{k}$$

We already know that the proof is wrong

$\Rightarrow$ this equivalence must be wrong as well.

Now we need to find why.

We have checked that the equivalence

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1} \Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \text{ holds}$$

However, the student adds $\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \leq \frac{k}{\sqrt{k}}$ which is not an equivalent transformation

Let us consider what happens when $k=1$

We want to prove the statement holds for $k+1$ ($k+1=2$)

We use the equivalences given by the student

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} \leq \sqrt{2} \Leftrightarrow 1 \leq \sqrt{2} - \frac{1}{\sqrt{2}} \quad (\text{so far correct})$$

$$\Leftrightarrow 1 \leq \frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}} \text{ (so far correct)}$$

$$\Leftrightarrow 1 \leq \frac{1}{\sqrt{2}} \text{ (so far correct)}$$

→ from here we can easily see that the statement doesn't hold

However the student continues and claims

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} \leq \sqrt{2} \Leftrightarrow 1 \leq \sqrt{2} - \frac{1}{\sqrt{2}}$$

$$\Leftrightarrow 1 \leq \frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$

$$\Leftrightarrow 1 \leq \frac{1}{2} \leq \frac{1}{\sqrt{1}} = \sqrt{1} = 1$$

↓
this is not an equivalent transformation

What we mean by the transformation not being equivalent is that in this exercise we have

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1} \Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}}, \text{ which}$$

implies But is not equivalent to

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}}$$

⇒ correct would be to say

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1} \Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k+1} - \frac{1}{\sqrt{k+1}}$$

$$\Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k+1}{\sqrt{k+1}} - \frac{1}{\sqrt{k+1}}$$

$$\Leftrightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}}$$

$$\Rightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \frac{k}{\sqrt{k+1}} \leq \frac{k}{\sqrt{k}}$$

$$\Rightarrow \frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k}$$

With the proof written as it is the student show that if:

1) $\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1}$ is true, and

2) $\frac{k}{\sqrt{k+1}} \leq \frac{k}{\sqrt{k}}$ is true, then

$$\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} \leq \sqrt{k}$$

However the implication can be used in only 1 direction

$\Rightarrow$ we cannot use the assumption from the Induction Hypothesis to show $\frac{1}{\sqrt{1}} + \dots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq \sqrt{k+1}$

Comment: The above solution is quite long and at times informal. It contains a lot of explanations that are aimed to make understanding the issue with the equivalence easier to and therefore is not exactly what would be expected from you on the exam and from the homework.

Steps we made to solve this exercise:

1. First we checked whether the claim was actually true.

!Be careful! It is possible for the solution to have been wrong even if the claim was true.

2. Since the mistake was not immediately obvious, we started by checking what was correct, so that we can find the place the mistake occurs by exclusion.

3. After we found where the mistake occurs we gave an example with an exact value of k to understand what exactly goes wrong.

Sometimes it might be very hard to make the calculations for the exact value of k , then it might be better to not try to give an example. The example was more for our understanding and you may not include it in your final solution.

Exercise 1.3 Asymptotic growth (1 point).

Recall the concept of asymptotic growth that we introduced in Exercise sheet 0: If $f, g : \mathbb{N} \rightarrow \mathbb{R}^+$ are two functions, then:

- We say that f grows asymptotically slower than g if $\lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = 0$. If this is the case, we also say that g grows asymptotically faster than f .

Prove or disprove each of the following statements.

(a) $f(m) = 100m^3 + 10m^2 + m$ grows asymptotically slower than $g(m) = 0.001 \cdot m^5$.

(b) $f(m) = \log(m^3)$ grows asymptotically slower than $g(m) = (\log m)^3$.

(c) $f(m) = e^{2m}$ grows asymptotically slower than $g(m) = 2^{3m}$.

Hint: Recall that for all $n, m \in \mathbb{N}$, we have $n^m = e^{m \ln n}$.

(d)* If $f(m)$ grows asymptotically slower than $g(m)$, then $\log(f(m))$ grows asymptotically slower than $\log(g(m))$.

(e)* $f(m) = \ln(\sqrt{\ln(m)})$ grows asymptotically slower than $g(m) = \sqrt{\ln(\sqrt{m})}$.

Hint: You can use L'Hôpital's rule from sheet 0.

Solution:

In order to determine if $f(m)$ grows asymptotically slower than $g(m)$ we will check if $\lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = 0$

$$\begin{aligned} \text{(a)} \quad \lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} &= \frac{100m^3 + 10m^2 + m}{0.001 \cdot m^5} = \lim_{m \rightarrow \infty} \frac{100}{10^{-3} \cdot m^2} + \frac{10}{10^{-3} m^3} + \frac{1}{10^{-3} m^4} \\ &= \lim_{m \rightarrow \infty} \frac{10^5}{m^2} + \frac{10^4}{m^3} + \frac{10^3}{m^4} = 0 \Rightarrow \text{the statement is true} \end{aligned}$$

$$\text{(b)} \quad \lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = \lim_{m \rightarrow \infty} \frac{\log(m^3)}{(\log m)^3} = \lim_{m \rightarrow \infty} \frac{3 \log(m)}{\log^3 m} = \lim_{m \rightarrow \infty} \frac{3}{\log^2 m} = 0$$

$\Rightarrow$ the statement is true

$$\text{(c)} \quad \lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = \lim_{m \rightarrow \infty} \frac{e^{2m}}{2^{3m}} = \lim_{m \rightarrow \infty} \frac{e^{2m}}{e^{3m \ln 2}} = \lim_{m \rightarrow \infty} e^{2m - 3m \ln 2}$$

$$= \lim_{m \rightarrow \infty} e^{(2 - 3 \ln 2)m} = \lim_{m \rightarrow \infty} e^{(-0.029) \cdot m} = 0 \Rightarrow \text{the statement is true}$$

(d) We will disprove the statement by giving counterexample of 2 functions for which it does not hold

Let $f(m) = m^2$ and $g(m) = m^3$

First, we will show that $f(m)$ grows asymptotically slower than $g(m)$

$$\lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = \lim_{m \rightarrow \infty} \frac{m^2}{m^3} = \lim_{m \rightarrow \infty} \frac{1}{m} = 0$$

$\Rightarrow f(m)$ grows asymptotically slower than $g(m)$

Now we will prove that $\log(f(m))$ does not grow asymptotically slower than $\log(g(m))$.

$$\lim_{m \rightarrow \infty} \frac{\log(f(m))}{\log(g(m))} = \lim_{m \rightarrow \infty} \frac{\log(m^2)}{\log(m^3)} = \lim_{m \rightarrow \infty} \frac{2 \log(m)}{3 \log(m)} = \frac{2}{3}$$

Since $\lim_{m \rightarrow \infty} \frac{\log(f(m))}{\log(g(m))} = \frac{2}{3} \neq 0$ this means that $\log(f(m))$ does

not grow asymptotically slower than $\log(g(m))$

$\Rightarrow$ the statement is false.

$$(e) \lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = \lim_{m \rightarrow \infty} \frac{\ln(\sqrt{\ln(m)})}{\sqrt{\ln(m)}} = \lim_{m \rightarrow \infty} \frac{\ln(\sqrt{\ln(m)})}{\sqrt{\ln(m^{\frac{1}{2}})}}$$

$$= \lim_{m \rightarrow \infty} \frac{\ln(\sqrt{\ln(m)})}{\sqrt{\frac{1}{2} \ln(m)}} = \lim_{m \rightarrow \infty} \frac{\ln(\sqrt{\ln(m)})}{\frac{1}{\sqrt{2}} \sqrt{\ln(m)}} = \lim_{m \rightarrow \infty} \frac{\sqrt{2} \ln(\sqrt{\ln(m)})}{\sqrt{\ln(m)}}$$

Let $y = \sqrt{\ln(m)}$.

We know that when $m \rightarrow \infty$, then $y \rightarrow \infty$

$$\Rightarrow \lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = \lim_{m \rightarrow \infty} \frac{\sqrt{2} \ln(\sqrt{\ln(m)})}{\sqrt{\ln(m)}} = \lim_{y \rightarrow \infty} \frac{\sqrt{2} \ln y}{y}$$

$$= \lim_{y \rightarrow \infty} \frac{(\sqrt{2} \ln y)'}{y'} = \lim_{y \rightarrow \infty} \frac{\sqrt{2} \cdot \frac{1}{y}}{1} = \lim_{y \rightarrow \infty} \frac{\sqrt{2}}{y} = 0$$

d'Hopital's rule

$$\Rightarrow \lim_{m \rightarrow \infty} \frac{f(m)}{g(m)} = 0 \Rightarrow \text{the statement is true}$$

(*) You are not expected to calculate that by hand.

Comment: The master solution for part (d) uses a different counterexample, but the overall idea is the same. The master solution for part (e) uses d'Hopital's rule twice instead of substitution.

This solution demonstrates an appropriate level of detail and

formalism for the exam.

Exercise 1.4 Proving Inequalities.

(a) Prove the following inequality by mathematical induction

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n} \leq \frac{1}{\sqrt{3n+1}}, \quad n \geq 1.$$

In your solution, you should address the base case, the induction hypothesis and the induction step.

(b)* Replace $3n + 1$ by $3n$ on the right side, and try to prove the new inequality by induction. This inequality is even weaker, hence it must be true. However, the induction proof fails. Try to explain to yourself how is this possible?

However, as argued above in the exercise statement, the inequality is still true. We are just not able to prove it directly via mathematical induction.

Solution:

Base case: $n = 1$

$$\frac{1}{2} \leq \frac{1}{2} = \frac{1}{\sqrt{3 \cdot 1 + 1}}$$

$\Rightarrow$ The statement is true for $n = 1$

Hypothesis: We assume the statement holds for $n = k, k \geq 1$

$$\Rightarrow \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2k-1}{2k} \leq \frac{1}{\sqrt{3k+1}}$$

Induction step: We will show the statement holds for $n = k+1, k \geq 1$

$$\text{We want to prove } \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)+1}}$$

From the Induction Hypothesis we know

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2k-1}{2k} \leq \frac{1}{\sqrt{3k+1}}$$

We multiply both sides of the inequality by $\frac{2(k+1)-1}{2(k+1)}$ and since $k \geq 1$ and thus $\frac{2(k+1)-1}{2(k+1)} > 0$, we get:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2k-1}{2k} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)}$$

Now we will show that:

$$\frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)+1}} \quad (*)$$

$$\frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)+1}}$$

$$\Leftrightarrow \frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3k+4}}$$

$$\Leftrightarrow \frac{(2(k+1)-1)}{2(k+1)} \leq \frac{\sqrt{3k+1}}{\sqrt{3k+4}} \quad (k \geq 1 \Rightarrow \sqrt{3k+1} > 0)$$

$$\Leftrightarrow \left(\frac{2k+1}{2k+2} \right)^2 \leq \frac{3k+1}{3k+4} \quad (\text{Both sides were non-negative})$$

$$\Leftrightarrow \frac{4k^2+4k+1}{4k^2+8k+4} \leq \frac{3k+1}{3k+4}$$

$$\Leftrightarrow (4k^2+4k+1)(3k+4) \leq (3k+1)(4k^2+8k+4) \quad \left(\begin{array}{l} 4k^2+8k+4 > 0 \text{ and} \\ 3k+4 > 0 \end{array} \right)$$

$$\Leftrightarrow 12k^3+12k^2+3k+16k^2+16k+4 \leq 12k^3+24k^2+12k+4k^2+8k+4$$

$$\Leftrightarrow 12k^3+28k^2+19k+4 \leq 12k^3+28k^2+20k+4 \quad | -(12k^3+28k^2+4)|$$

$$\Leftrightarrow 19k \leq 20k \quad | -19k$$

$$\Leftrightarrow 0 \leq k, \text{ which holds since we know } k \geq 1$$

$$\Rightarrow \frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)+1}} \quad \text{holds}$$

$\Rightarrow$ We have shown:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2k-1}{2k} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)}$$

$$\text{and } \frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)+1}}$$

$$\Rightarrow \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3k+1}} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)+1}}$$

$$\Rightarrow \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)+1}}$$

$\Rightarrow$ the statement holds for $n=k+1$

$\Rightarrow$ By the principle of mathematical induction, we conclude the inequality is true for any positive integer n .

(*) We advice against using equivalences in the Induction step in order to avoid imstakes as in Exercise 1.2. However, in this case we are not using equivalences to directly show that the statement we are trying to prove is equivalent to statement we already know is true. As you can see in the master solutions, the equivalence is carried out as a proof to a lemma, which is then used in the induction step. In such cases is OK to use equivalences.

(6) We will try to prove the statement By mathematical induction

Base case: $n = 1$

$$\frac{1}{2} \leq \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3 \cdot 1}} \quad \checkmark$$

Hypothesis: We assume the statement holds for $n = k, k \geq 1$

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2k-1}{2k} \leq \frac{1}{\sqrt{3k}}$$

Induction step: We will show the statement holds for $n = k+1, k \geq 1$

We want to show:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2k-1}{2k} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)}}$$

From the Induction Hypothesis we know that:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2k-1}{2k} \leq \frac{1}{\sqrt{3k}}$$

Since $k \geq 1$ and thus $\frac{2(k+1)-1}{2(k+1)} > 0$ we can multiply by

$\frac{2(k+1)-1}{2(k+1)}$ on both sides and we get:

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \dots \cdot \frac{2k-1}{2k} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3k}} \cdot \frac{2(k+1)-1}{2(k+1)}$$

Analogous to part (a) we are going to try to prove that

$$\frac{1}{\sqrt{3k}} \cdot \frac{2(k+1)-1}{2(k+1)} \leq \frac{1}{\sqrt{3(k+1)}}$$

$$\Leftrightarrow \frac{2(k+1)-1}{2(k+1)} \leq \frac{\sqrt{3k}}{\sqrt{3(k+1)}} \quad (\sqrt{3k} > 0)$$

$$\Leftrightarrow \left(\frac{2k+1}{2k+2} \right)^2 \leq \frac{3k}{3(k+1)} \quad (\text{Both sides were non-negative})$$

$$\Leftrightarrow \frac{4k^2+4k+1}{4k^2+8k+4} \leq \frac{3k}{3(k+1)}$$

$$\Leftrightarrow (4k^2+4k+1)(3k+3) \leq (4k^2+4k+1)(3k)$$

$$\Leftrightarrow 12k^3+24k^2+15k+3 \leq 12k^3+24k^2+12k$$

$$\Leftrightarrow 3k+3 \leq 0$$

Which does not hold, since $k \geq 1$.

The issue occurs in the induction hypothesis. The inequality we assume is weaker than the inequality in part a). This means that it gives a larger upper bound for the expression on the left-hand side. This bound is too large and makes it impossible to do the induction step.

Comment: For part (a) the master solution follows a different order (i.e. the inequality (*) is carried out as a lemma and is proven before the start of the induction).

This solution demonstrates the appropriate level of detail and formalism for the exam.

Asymptotic notation (O-notation)

Notation: $N = \{n_0, n_0+1, \dots\}$ - the set of all possible arguments of the function f . (In our case we consider f to be the function that gives us the runtime of

an algorithm and therefore N is a set of all possible input sizes.)! N is not necessarily $\mathbb{N}$ (the set of the natural numbers). It is possible that $N = \mathbb{N}$, but this is not always true. N is just a set of all possible inputs of f .

Definition: For $f: N \rightarrow \mathbb{R}^+$, we define
 $O(f) := \{g: N \rightarrow \mathbb{R}^+ \mid \exists C \in \mathbb{R}^+ : \forall n \in N : g(n) \leq C \cdot f(n)\}$

Definition (with words) For a function f with domain N (domain are the possible values of the argument of f) and codomain $\mathbb{R}^+$ (codomain is the set of values of $f(x)$, $x \in N$) ($\mathbb{R}^+$ are the positive real numbers) we define $O(f)$. $O(f)$ is defined as the set of all functions g with domain N and codomain $\mathbb{R}^+$ for which there exists a positive real constant C , such that for every n in the domain N it holds: $g(n) \leq C \cdot f(n)$ (the value g takes in point n is less or equal to the value of f in the same point n multiplied by the constant C).

!Important: the constant C is the same for all $n \in N$

? Why do we consider functions $f: N \rightarrow \mathbb{R}^+$, where N is usually a subset of $\mathbb{N}$ and $f(n) \in \mathbb{R}^+$, $n \in N$?

Because we are trying to describe the runtime of an algorithm based on the input size. The input size is usually a natural number and the runtime cannot possibly be 0 or less.

? Why do we use N instead of $\mathbb{N}$?

Because sometimes for smaller values of n it is not always true that $g(n) \leq C \cdot f(n)$. Since the O -notation is asymptotic, the smaller values of n do not matter. Important is what happens as n goes to ∞ .

Theorem 1. Let N be an infinite subset of $\mathbb{N}$ and $f : N \rightarrow \mathbb{R}^+$ and $g : N \rightarrow \mathbb{R}^+$.

- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$, then $f \leq O(g)$ and $g \not\leq O(f)$.
- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = C \in \mathbb{R}^+$, then $f \leq O(g)$ and $g \leq O(f)$.
- If $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$, then $f \not\leq O(g)$ and $g \leq O(f)$.

The following theorem can also be helpful when working with O -notation.

Theorem 2. Let $f, g, h : N \rightarrow \mathbb{R}^+$. If $f \leq O(h)$ and $g \leq O(h)$, then

1. For every constant $c > 0$, $c \cdot f \leq O(h)$.
2. $f + g \leq O(h)$.

Notice that for all real numbers $a, b > 1$, $\log_a n = \log_a b \cdot \log_b n$ (where $\log_a b$ is a positive constant). Hence $\log_a n \leq O(\log_b n)$. So you don't have to write bases of logarithms in asymptotic notation, that is, you can just write $O(\log n)$. Be careful though, as this does not apply when the logarithms are in some exponent! For example, $e^{\ln(n)} = n$ but $e^{\log(n)} = n^{\log e}$, and these two expressions are certainly not of the same order.

Moodle Quiz

For $n \in \mathbb{N}$, let $f(n) = n^2 + 1001n + n^3$ and $g(n) = 10n^3$. Then $f(n)$ has the same asymptotic growth rate as $g(n)$ (meaning that $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \in \mathbb{R}_+$).

- ☒ True
☐ False

Solution: We calculate $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} &= \lim_{n \rightarrow \infty} \frac{n^2 + 1001n + n^3}{10n^3} = \lim_{n \rightarrow \infty} \frac{1}{10n} + \frac{1001}{10n^2} + \frac{1}{10} \\ &= \frac{1}{10} \in \mathbb{R}^+ \Rightarrow \text{the statement is true.} \end{aligned}$$

$$n^4 \leq O\left(\frac{n^4}{\log(n)}\right)$$

- ☐ True
☒ False

Solution: We know from Theorem 1, that

$$n^4 \leq O\left(\frac{n^4}{\log(n)}\right) \text{ if } \lim_{n \rightarrow \infty} \frac{\frac{n^4}{\log(n)}}{\frac{n^4}{\log(n)}} = 0 \text{ or } C, C \in \mathbb{R}^+$$

We calculate $\lim_{n \rightarrow \infty} \frac{n^4}{\frac{n^4}{\log(n)}}$

$$\lim_{n \rightarrow \infty} \frac{n^4}{\frac{n^4}{\log(n)}} = \lim_{n \rightarrow \infty} \log(n) = \infty \Rightarrow \text{from Th. 1 we know } n^4 \neq O\left(\frac{n^4}{\log(n)}\right)$$

$$e^{3 \ln(n)} \leq O(n^2)$$

☐ True

☒ False

We want to use Th. 1 so we calculate $\lim_{n \rightarrow \infty} \frac{e^{3 \ln(n)}}{n^2}$

$$\lim_{n \rightarrow \infty} \frac{e^{3 \ln(n)}}{n^2} = \lim_{n \rightarrow \infty} \frac{n^3}{n^2} = \lim_{n \rightarrow \infty} n = \infty \Rightarrow \text{from Th. 1 we know } e^{3 \ln(n)} \neq O(n^2)$$

Let $f(n) = 6n^2 + 5n + 10$ (for $n \in \mathbb{N}$). For each of the following definitions of $g(n)$, is $g(n) \leq O(f(n))$?

True	False	
☒	☐	$g(n) = 123456789n - 200\sqrt{n}$
☐	☒	$g(n) = 0.01n^2 \log(n)$
☐	☒	$g(n) = 10n^3 + 5n + 1000$

Scoring method: Subpoints

We will use Th. 1. $\Rightarrow$ we calculate $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)}$

$$g(n) = 123456789n - 200\sqrt{n}$$

$$\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{123456789n - 200\sqrt{n}}{6n^2 + 5n + 10}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 \left(123456789 \cdot \frac{1}{n} - 200 \cdot \frac{1}{n\sqrt{n}} \right)}{n^2 \left(6 + \frac{5}{n} + \frac{10}{n^2} \right)} = 0 \Rightarrow \text{from Th. 1 we know } g(n) \leq O(f(n))$$

$$g(n) = 0.01n^2 \log(n)$$

$$\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{0.01 \cdot n^2 \log(n)}{6n^2 + 5n + 10} = \lim_{n \rightarrow \infty} \frac{n^2 (0.01 \cdot \log(n))}{n^2 \left(6 + \frac{5}{n} + \frac{10}{n^2} \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{0.01 \cdot \log(n)}{6} = \infty \Rightarrow \text{from Th. 1 we know } g(n) \neq O(f(n))$$

$$g(n) = 10n^3 + 5n + 100$$

$$\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = \lim_{n \rightarrow \infty} \frac{10n^3 + 5n + 100}{6n^2 + 5n + 10} = \lim_{n \rightarrow \infty} \frac{n^2(10n + \frac{5}{n} + 100 \cdot \frac{1}{n^2})}{n^2(6 + \frac{5}{n} + \frac{10}{n^2})}$$

$$= \lim_{n \rightarrow \infty} \frac{10n}{6} = \infty \Rightarrow \text{from Th. 1 we know } g(n) \neq O(f(n))$$

Let $f: \mathbb{N} \rightarrow \mathbb{R}_+$ be some function for which we would like to prove $f(n) \geq n^2$ for every $n \geq 1$. Assume that you have proven that:

- $f(2) \geq 2^2$
- If $f(k) \geq k^2$ holds for an arbitrary positive integer k , then $f(k+1) \geq (k+1)^2$ holds.

Then, $f(n) \geq n^2$ holds for all positive integers $n \geq 1$.

☐ True

☒ False

Solution: The statement that $f(n) \geq n^2$ holds for all positive integers $n \geq 1$ is false, since the Base case is for $n=2$ and with the induction step the claim is proved for values of n bigger than 2. $\Rightarrow$ It is never shown that the inequality holds for $n=1$.

Comment: Those solutions to the Moodle Quiz questions are quite formal and are not expected during the Quiz. For most of the above questions it is sufficient to just compare the dominating terms of the two functions. It is possible that in future Moodle Quizzes a short computation will be needed.

Exercises with O-notation
Level 1.

Claim	True	False
1) $f(n) = 3n + 5, g(n) = n^2, f \leq O(g)$	✓	
2) $f(n) = 7n^2 + 4n + 1, g(n) = n^2, f \leq O(g)$	✓	
3) $f(n) = n^2 + \log n, g(n) = n^3, f \neq O(g)$		✓
4) $f(n) = 50 \log n, g(n) = n, f \leq O(g)$	✓	
5) $f(n) = n \log n, g(n) = n^{1.5}, f \neq O(g)$		✓

Level 2

	Claim	True	False
1)	$f(n) = n^3, g(n) = 2^n, f \leq O(g)$	✓	
2)	$f(n) = 2^n, g(n) = 3^n, f \leq O(g)$	✓	
3)	$f(n) = \sqrt{n}, g(n) = \ln^2 n, f \leq O(g)$		✓
4)	$f(n) = n \log(n), g(n) = n \cdot (\log n)^2, f \leq O(g)$	✓	
5)	$f(n) = n^{\ln(n)}, g(n) = 2^n, f \leq O(g)$	✓	

Solution: 1) $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^3}{2^n} = \lim_{n \rightarrow \infty} \frac{e^{3 \ln n}}{e^{n \ln 2}} = \lim_{n \rightarrow \infty} e^{3 \ln n - n \ln 2}$

$= e^{\lim_{n \rightarrow \infty} 3 \ln n - n \ln 2} = e^{-\infty} = 0 \Rightarrow \text{true}$

$\lim_{n \rightarrow \infty} 3 \ln n - n \ln 2 = \lim_{n \rightarrow \infty} n \left(\frac{3 \ln n}{n} - \ln 2 \right) = \lim_{n \rightarrow \infty} (-\ln 2) \cdot n = -\infty$

3) $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\ln^2 n} = \lim_{n \rightarrow \infty} \frac{(\sqrt{n})'}{(\ln^2 n)'} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2} \frac{1}{\sqrt{n}}}{2 \ln n \cdot \frac{1}{n}}$

$= \lim_{n \rightarrow \infty} \frac{\frac{1}{2} \frac{n}{\sqrt{n}}}{2 \ln n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{4 \ln n} = \lim_{n \rightarrow \infty} \frac{(\sqrt{n})'}{(4 \ln n)'} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2} \frac{1}{\sqrt{n}}}{4 \cdot \frac{1}{n}}$

$= \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{8} = \infty \Rightarrow \text{false}$

5) $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^{\ln(n)}}{2^n} = \lim_{n \rightarrow \infty} \frac{e^{\ln(n) \cdot \ln(n)}}{e^{n \ln 2}}$

$= \lim_{n \rightarrow \infty} e^{\ln^2(n) - n \ln 2} = e^{\lim_{n \rightarrow \infty} \ln^2(n) - n \ln 2} = e^{-\infty} = 0 \Rightarrow \text{true}$

$\lim_{n \rightarrow \infty} \ln^2(n) - n \ln 2 = \lim_{n \rightarrow \infty} n \left(\frac{\ln^2(n)}{n} - \ln 2 \right) \stackrel{(*)}{=} \lim_{n \rightarrow \infty} n (-\ln 2) = -\infty$

$(*) \lim_{n \rightarrow \infty} \frac{\ln^2(n)}{n} = \lim_{n \rightarrow \infty} \frac{(\ln^2(n))'}{n'} = \lim_{n \rightarrow \infty} \frac{2 \ln(n) \cdot \frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{2 \ln(n)}{n}$

$= \lim_{n \rightarrow \infty} \frac{(2 \ln(n))'}{n'} = \lim_{n \rightarrow \infty} \frac{2 \cdot \frac{1}{n}}{1} = \lim_{n \rightarrow \infty} \frac{2}{n} = 0$

Level 3 (Try yourselves)

	Claim	True	False
1)	$f(n) = n^{1.1}$, $g(n) = n \log n$, $f \leq O(g)$		
2)	$f(n) = 2^{\frac{n}{\ln n}}$, $g(n) = 2^{n^{0.9}}$, $f \leq O(g)$		
3)	$f(n) = n^{e^n}$, $g(n) = e^{\sqrt{n}}$, $f \leq O(g)$		

Mathematical Induction Exercises

Exercise 1 The sequence $\{a_n\}$ is given with the recursive formula:

$$a_1 = 1$$

$$a_n = \frac{a_{n-1}}{a_{n-1} + 1}$$

Find a formula for a_n in terms of n .

Exercise 2 Prove the equality for $\forall n \in \mathbb{N}$

$$2 + 4 + 15 + \dots + (n^2 + 2n - 1) = \frac{n(2n^2 + 9n + 1)}{6}$$

Exercise 3 Prove the inequality for $\forall n \in \mathbb{N}$

$$\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{n+n} > \frac{13}{24}$$